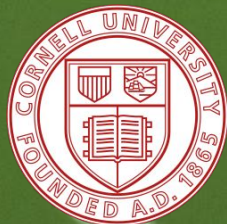


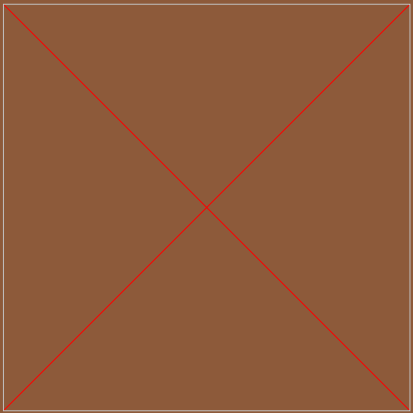
# Introduction to eBird and its Data

Fabiola Rodríguez Vásquez  
*Cornell University*  
fgr23@cornell.edu

Aimee Van Tatenhove  
*Cornell Lab of Ornithology*  
amv48@cornell.edu



Prairie Warbler by Luke Seitz, Macaulay Library



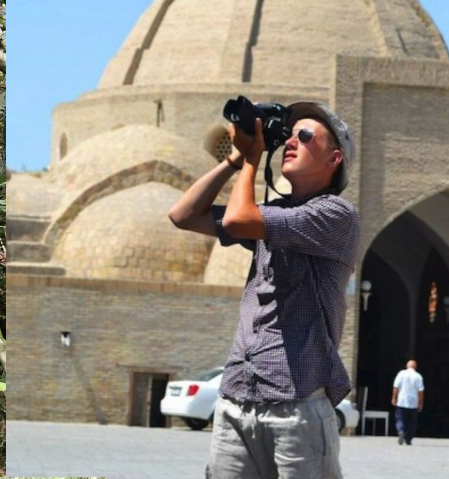
# What is eBird?

*¿Qué es eBird?*

# eBird

is an online database of **bird observations** collected by **participatory scientists**, providing data about the global **distribution and abundance** of bird species.

es una base de datos de **registros de aves** colectada por participantes de la comunidad, dando datos sobre distribución global y **abundancia de especies**.



# eBird Overview

*Descripción general de eBird*

>2 billion bird observations

*>2 mil millones de registros*

All countries of the world

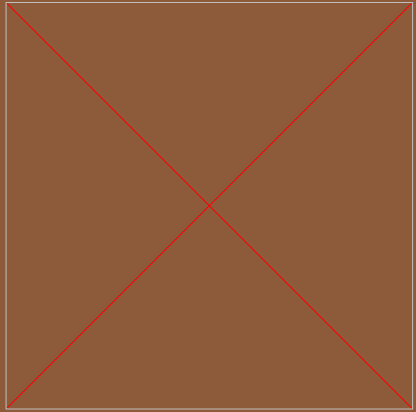
*Todos los países del mundo*

10 827 especies

250 peer reviewed publications in the last 5 years

*250 publicaciones revisadas por pares en los últimos cinco años*

2002



What is the  
structure of an  
eBird dataset?

*¿Cuál es la estructura de un  
conjunto de datos de eBird?*

Mon 2 Jul 2018 1:11 PM

[Edit Date and Effort](#)

Winter Trail, Alaska, US (71.251, -156.491)

[North Slope County, Alaska, United States](#)[Edit Location](#)

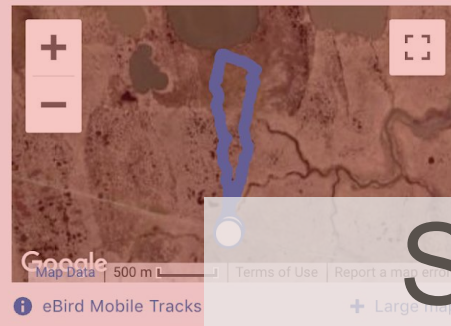
Matthew Strimas-Mackey

[Share](#)Andrew Spencer , Ian Davies ,  
Nathan Pieplow Traveling Complete 

Are you submitting a **complete checklist** of the birds you were able to identify? **Yes**

[Learn more](#)

4 1 hr, 50 min 2.37 mi



Submitted from eBird Android, version 1.9

[Edit comments](#)

SUBMIT ANOTHER FOR...

Same location and date

Winter Trail, Alaska, US (71.251, -156.491), North Slope County, Alaska, US on Mon Jul 02, 2018

Same location

Winter Trail, Alaska, US (71.251, -156.491), North Slope County, Alaska, US

Same area and date

Another location near Winter Trail, Alaska, US (71.251, -156.491), North Slope County, Alaska, US on Mon Jul 02, 2018

Same area

Another location near Winter Trail, Alaska, US (71.251, -156.491), North Slope County, Alaska, US

Same date

Mon Jul 02, 2018

Different location and date

MEDIA POWERED BY [MACAULAY LIBRARY](#)

23

Species observed  
+1 other taxa

2

Species with photos

150 Greater White-fronted Goose

1 Tundra Swan

15 Northern Pintail

1 Spectacled Eider

BREEDING CODE: NE Nest with Eggs (Confirmed)

3 King Eider

25 Long-tailed Duck

120 American Golden-Plover

Presumably fully grown

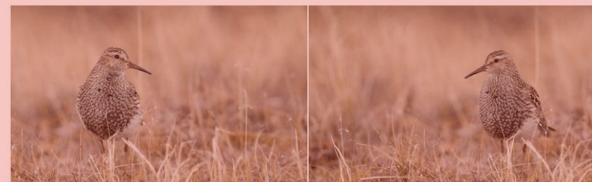
2 Dunlin

BREEDING CODE: NE Nest with Eggs (Confirmed)



20 Pectoral Sandpiper

BREEDING CODE: NE Nest with Eggs (Confirmed)



# Semistructured data

*Datos semiestructurados*

Tue 9 May 2023 10:41 AM  
 9030915, Pudahuel CL-Santiago (-33.4272,-70.7826) 📍 Pudahuel County, Región Metropolitana de Santiago

Tom Auer  
 Cullen Hanks 📷, Jenna Curtis 📷

Incidental Incomplete ⓘ

👤 3

Submitted from eBird for iOS, version 2.17.3

🌿 3 Species observed

|   |                              |
|---|------------------------------|
| X | Rock Pigeon (Feral Pigeon) * |
| 2 | Southern Lapwing             |
| 5 | Austral Thrush               |

CHECKLIST 547044589  
 Mon 2 Jul 2018 1:11 PM [Edit Date and Time](#)  
 Winter Trail, Alaska, US (71.251, -156.491) 📍 North Slope County, Alaska, United States [Edit Location](#)

Matthew Strimas-Mackey [Share](#)

Andrew Spencer 📷, Ian Davies 📷, Nathan Pappano 📷

Traveling Complete 🟢  
 Are you submitting a complete checklist of the birds you were able to identify? Yes [Learn more](#)

📍 4 🕒 1 hr, 50 min 📏 2.37 mi

 [eBird Mobile Tracks](#) [+ Large map](#)

Submitted from eBird Android, version 1.9 [Edit comments](#)

SUBMIT ANOTHER FOR...

Same location and date  
 Winter Trail, Alaska, US (71.251, -156.491), North Slope County, Alaska, US on Mon Jul 02, 2018

Same location  
 Winter Trail, Alaska, US (71.251, -156.491), North Slope County, Alaska, US

Same area and date  
 Another location near Winter Trail, Alaska, US (71.251, -156.491), North Slope County, Alaska, US on Mon Jul 02, 2018

Same area  
 Another location near Winter Trail, Alaska, US (71.251, -156.491), North Slope County, Alaska, US

Same date  
 Mon Jul 02, 2018

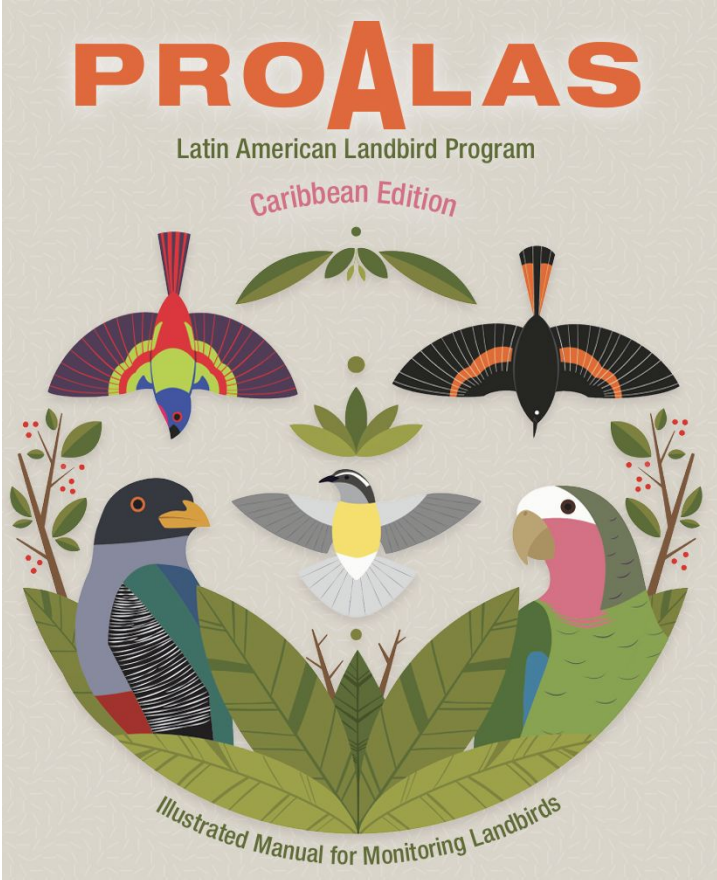
Different location and date

MEDIA POWERED BY MACAULAY LIBRARY

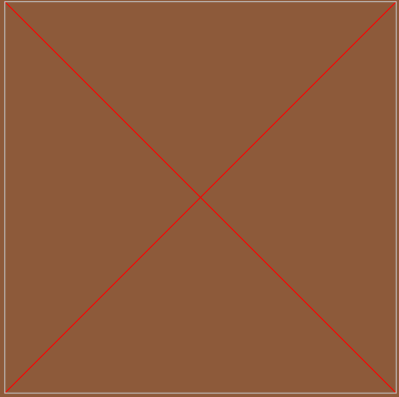
🌿 23 Species observed  
 +1 other taxa

📷 2 Species with photos

|     |                                                                                                                                                           |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 150 | Greater White-fronted Goose                                                                                                                               |
| 1   | Tundra Swan                                                                                                                                               |
| 15  | Northern Pintail                                                                                                                                          |
| 1   | Spectacled Eider<br>BREEDING CODE: NE Nest with Eggs (Confirmed)                                                                                          |
| 3   | King Eider                                                                                                                                                |
| 25  | Long-tailed Duck                                                                                                                                          |
| 120 | American Golden-Plover<br>Presumably failed brooders                                                                                                      |
| 12  | Dunlin<br>BREEDING CODE: NE Nest with Eggs (Confirmed)<br>             |
| 20  | Pectoral Sandpiper<br>BREEDING CODE: NE Nest with Eggs (Confirmed)<br> |

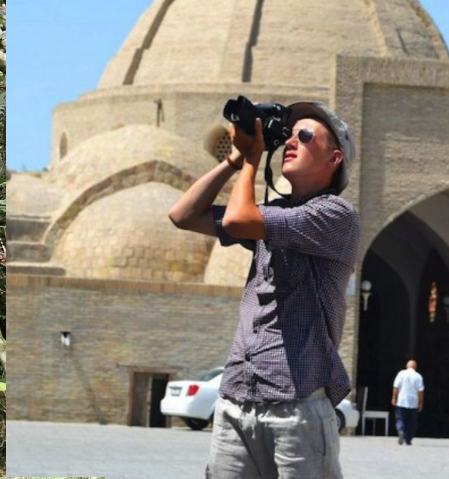


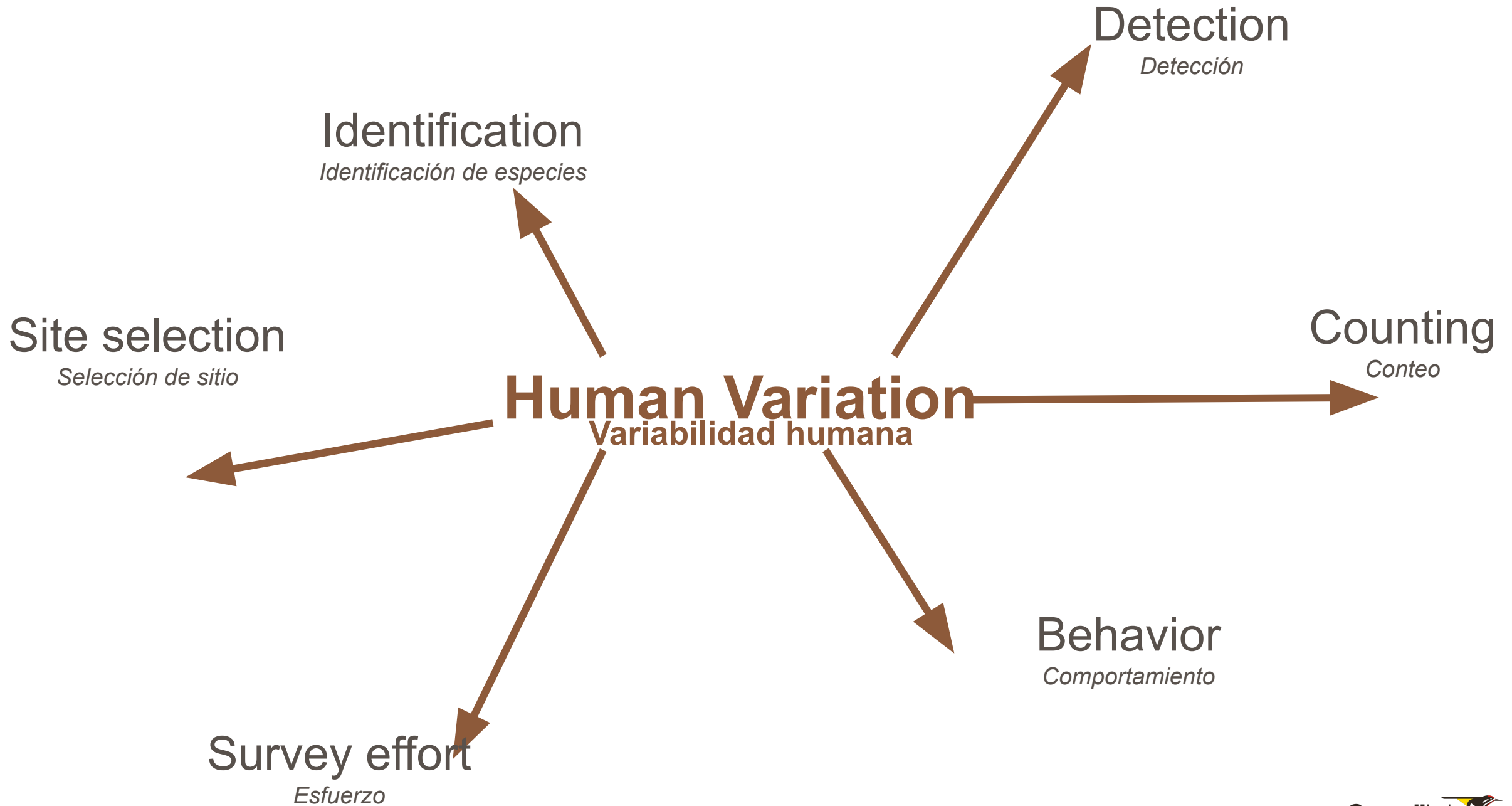
# Semi Dataset structure examples



# What to consider before using?

*¿Qué debo considerar antes  
de usar?*







# Human variations lead to these considerations:

*La variabilidad humana conduce a estas consideraciones:*

Taxonomic bias *Sesgo taxonómico*

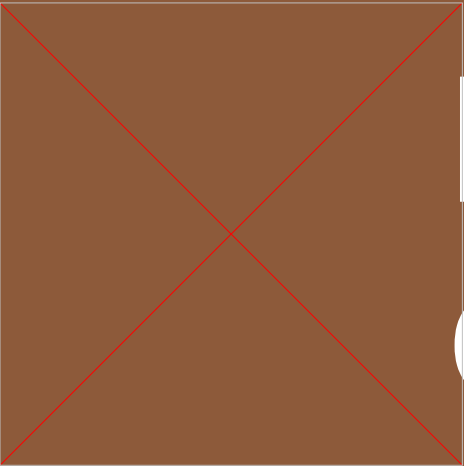
Spatial bias *Sesgo espacial*

Temporal bias *Sesgo temporal*

Class imbalance *Desbalance de clases*

Variation in sampling effort *Variación en esfuerzo*

Spatial precision *Precisión espacial*

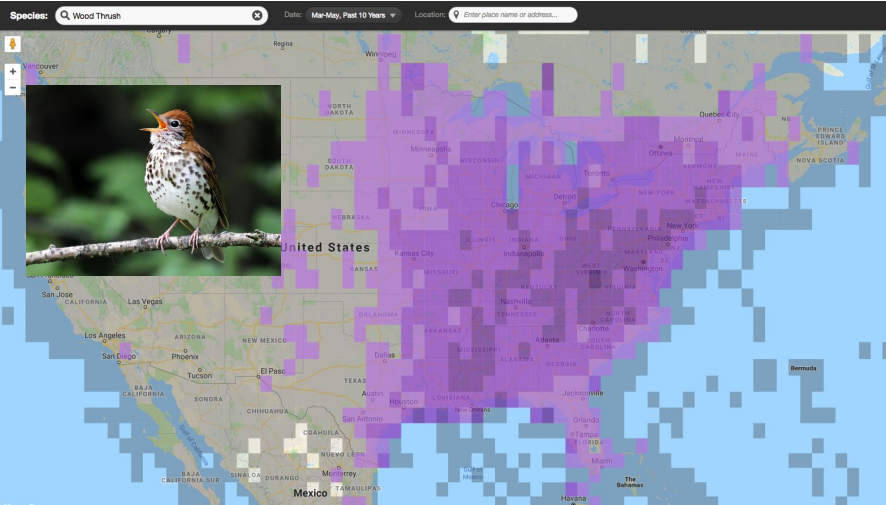


# How can I access eBird data?

*¿Cómo puedo acceder a los datos de eBird?*

# eBird website

(*página web eBird*)



Overview Recent Visits Illustrated Checklist

About POWERED BY MACAULAY LIBRARY AND eBIRD

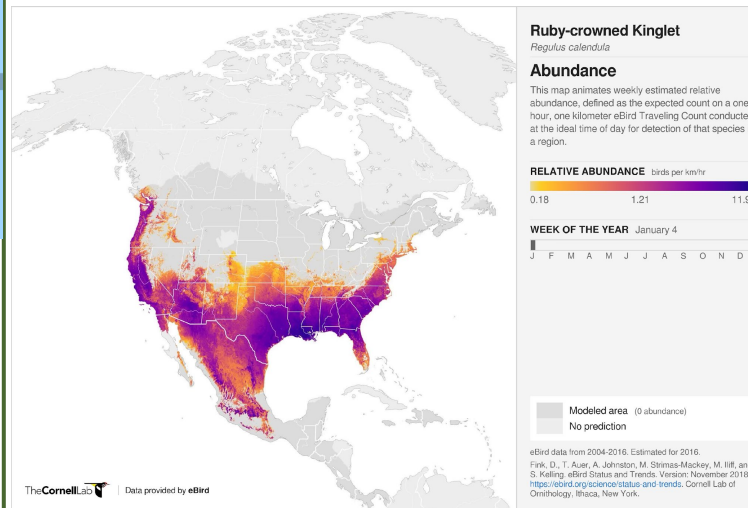
Jump to Species Enter species name All Media

234 Species observed 26 Species w/ photos 2 Species w/ audio 208 Species needing photos 232 Species needing audio



# eBird Status and Trends

(*eBird Estdados y Tendencias*)



# eBird raw data

(*eBird datos en crudo*)

auk 0.3.3 Vignette Function Reference

## auk: eBird Data Extraction and Processing in R

License GPL v3 Mac OSX & Linux passing Windows passing coverage 79%  
R Peer Reviewed CRAN 0.3.2 downloads 13K



## Overview

eBird is an online tool for recording bird observations. Since its inception, over 600 million records of bird sightings (i.e. combinations of location, date, time, and bird species) have been collected, making eBird one of the largest citizen science projects in history and an extremely valuable resource for bird research and conservation. The full eBird database is packaged as a text file and available for download as the **eBird Basic Dataset (EBD)**. Due to the large size of this dataset, it must be filtered to a smaller subset of desired observations before reading into R. This filtering is most efficiently done using AWK, a Unix utility and programming language for processing column formatted text data. This package acts as a front end for AWK, allowing users to filter eBird data before import into R.

## Installation

```
# cran release
install.packages("auk")

# or install the development version from github
# install.packages("remotes")
remotes::install_github("CornellLabofornithology/auk")
```

auk requires the Unix utility AWK, which is available on most Linux and Mac OS X machines. Windows users will first need to install **Cygwin** before using this package. Note that **Cygwin must be installed in the default location** (C:/cygwin/bin/gawk.exe or C:/cygwin64/bin/gawk.exe) in order for auk to work.

## Vignette

Full details on using auk to produce both presence-only and presence-absence data are outlined in the vignette, which can be accessed with vignette("auk").



**In this workshop we will  
use 'auk'**

***En este taller usaremos 'auk'***

**But first, questions?**

***Pero antes, preguntas?***

